

Assignment-007

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Batch-20

Design and Development of a Simple Personal Portfolio Decentralized Application (dApp) Frontend

Objective

To design and implement a simple decentralized application (dApp) frontend for a Personal Portfolio Landing Page that interacts with an Ethereum smart contract using HTML, CSS, JavaScript, and Web3.js.

The dApp will:

- Connect to the user's MetaMask wallet
- Read portfolio information (name, role, and skills) stored on the blockchain
- Display the data dynamically on a web page

Tools & Technologies Required

- VS Code (with Live Server extension recommended)
- Web browser (Chrome/Brave)
- MetaMask browser extension
- Node.js (optional, for local server)
- Ethereum test network (Sepolia / Ganache)
- Web3.js library

Prerequisites

- Basic knowledge of HTML, JavaScript
- Basic understanding of blockchain and smart contracts
- MetaMask wallet set up and connected to a test network

Html

```
<!DOCTYPE html>
<html>
<head>
<title>My Blockchain Portfolio</title>
<link rel="stylesheet" href="style.css">
<script src="https://cdn.jsdelivr.net/npm/web3@1.7.5/dist/web3.min.js"></script>
</head>
<body>
```

```
<h1>Personal Portfolio dApp</h1>
```

```
<button onclick="connectWallet()">Connect Wallet</button>
```

```
<div class="card">
<h2 id="name">Name:Mohan </h2>
<p id="role">Role:Student </p>
<p id="skills">Skills:python, DSA, Blockchain, Web3 </p>
```

```
<button onclick="loadPortfolio()">Load Portfolio</button>
</div>
```

```
<script src="script.js"></script>
```

```
</body>
</html>
```

Css

```
body{
```

```
font-family: Arial;
background:#f4f4f4;
text-align:center;
}

h1{
color:#333;
}
```

```
.card{
width:400px;
margin:auto;
background:white;
padding:20px;
border-radius:10px;
box-shadow:0 0 10px rgba(0,0,0,0.2);
}
```

```
button{
padding:10px 15px;
margin:10px;
background:#4CAF50;
color:white;
border:none;
border-radius:5px;
cursor:pointer;
}
```

```
button:hover{
background:#45a049;
}
```

Java script

```
let web3;
let contract;

const contractAddress = "PASTE_CONTRACT_ADDRESS";
```

```
const abi = [
{
  "inputs": [],
  "name": "getPortfolio",
  "outputs": [
    {"internalType": "string","name":"","type":"string"},
    {"internalType": "string","name":"","type":"string"},
    {"internalType": "string","name":"","type":"string"}
  ],
  "stateMutability": "view",
  "type": "function"
}
];
```

```
async function connectWallet(){
```

```
if(window.ethereum){
```

```
web3 = new Web3(window.ethereum);
```

```
await window.ethereum.request({method:'eth_requestAccounts'});
```

```

alert("Wallet Connected");

contract = new web3.eth.Contract(abi, contractAddress);

}

else{

alert("Install MetaMask");

}

}

async function loadPortfolio(){

const data = await contract.methods.getPortfolio().call();

document.getElementById("name").innerHTML="Name: "+data[0];
document.getElementById("role").innerHTML="Role: "+data[1];
document.getElementById("skills").innerHTML="Skills: "+data[2];

}

```

Output:

